MAPLE: Manifold-Activated Pruning of Latent Experts

Empirical Activation-Guided Expert Distillation for Mixture-of-Experts Language Models Below 2 Bits Per Weight

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Paper under double-blind review

Abstract

We present MAPLE (**M**anifold-**A**ctivated **P**runing of **L**atent **E**xperts), a post-training structural compression method for fine-grained Mixture-of-Experts (MoE) large language models that achieves sub-2-bit-per-weight effective parameter density while retaining high task-level semantic fidelity. In fine-grained MoE models trained on expansive multilingual corpora, a substantial fraction of routed expert capacity remains dormant for any specific deployment subdomain. We exploit this structural phenomenon by profiling the empirical expert activation manifold during native forward passes over a representative domain corpus. Each expert is scored via a composite metric combining activation frequency and input feature energy: $s_{e,l} = C_{e,l} \cdot \sqrt{\bar{\sigma}_{e,l} + \epsilon}$. Applied to Gemma 4 26B-A4B (128 routed experts, 30 layers, 9.92 GB in IQ2_XXS quantization), MAPLE excises 80 dormant experts per layer to assemble an optimized 48-expert container of 4.75 GB (effective 1.48 bits per parameter, 2.09× compression). The pruned model achieves 100% GPU VRAM residency on a consumer 6 GB laptop GPU (RTX 4050), accelerating generation by 6.57× from 6.9 to 45.4 tokens/second by eliminating PCIe memory-paging bottlenecks. Standardized evaluation across code synthesis, data structure implementation, scientific deduction, and formal logic demonstrates a 93.8% mean keyword fidelity retention. We document three subtle failure modes overcome during design: proxy-embedding routing estimation, cascaded normalization suppression, and DRY sampling interference.

1 Introduction

The proliferation of Mixture-of-Experts (MoE) architectures in frontier large language models [8, 2, 16] has introduced a structurally novel compression opportunity that conventional quantization methods fail to exploit. In a dense transformer, every weight participates in every forward pass; compression therefore operates uniformly across the weight space. In a MoE model, however, only a small fraction of the total expert capacity is activated per token—typically k experts out of E total, with $k/E \approx 0.03$–0.06 for modern fine-grained architectures. The question this paper asks is deceptively simple: *for a specific deployment subdomain, which k' out of E experts ever matter—and can we safely discard the rest entirely?*

The practical stakes are high. Consumer-grade GPUs have VRAM ceilings of 6–12 GB. The difference between a model that *fits entirely in VRAM* and one that *must page weights through CPU RAM* is not a minor quality difference; it is a qualitative cliff. In our empirical setting, the full-precision-routed Gemma 4 26B-A4B at 9.92 GB runs at 6.9 tokens per second with CPU paging on an RTX 4050 6GB. After MAPLE compression to 4.75 GB it runs at 45.4 tokens per second entirely within VRAM. The user experience—from barely interactive to real-time conversational—is transformed.

Prior work on MoE compression has largely focused on weight quantization [5, 11], structured pruning of attention heads [14], and static expert selection based on router logits [9]. What is missing is a *forward-pass empirical* approach that uses the model's own observed activation statistics, computed on a representative calibration corpus, to decide which experts survive compression. This is precisely what MAPLE provides.

This paper makes the following contributions:

1. We introduce the *empirical activation manifold* of a MoE model: the per-layer distribution of expert call counts and input energy statistics, obtained through native forward-pass profiling (§4).
2. We define the MAPLE expert scoring function $s_e = c_e \cdot \sqrt{\bar{\sigma}_e}$, where c_e is the empirical call count and $\bar{\sigma}_e$ is the mean input L2 energy of expert e, and show it outperforms count-only or energy-only selection (§4).
3. We document three distinct failure modes that invalidate naive compression approaches and explain precisely why each fails (§5).
4. We demonstrate 93.8% task fidelity retention, 2.09× compression, 1.48 effective BPW, and 6.57× throughput improvement on a commodity laptop GPU for a 26-billion-parameter MoE model (§6).
5. We release all code, calibration data, and expert selection maps openly in reproducible repositories.[1]

2 Background

2.1 Mixture-of-Experts Language Models

A Mixture-of-Experts transformer replaces the dense feed-forward network (FFN) in each transformer block with a set of E parallel expert networks and a learned router. For each input token hidden state $\mathbf{h} \in \mathbb{R}^d$, the router computes logits $\mathbf{g} = \mathbf{h}W_{\text{gate}} \in \mathbb{R}^E$, selects the top-$k$ experts by logit magnitude, applies a softmax-normalized weighting over the selected experts, and sums their outputs:

$$\text{FFN}_{\text{MoE}}(\mathbf{h}) = \sum_{e \in \text{top}_k(\mathbf{g})} \text{softmax}(\mathbf{g})_e \cdot \text{FFN}_e(\mathbf{h}), \tag{1}$$

where each FFN_e is a standard two-layer feed-forward network with gate, up-projection, and down-projection matrices.

Fine-grained MoE architectures [2] use large E (64–256 experts) with small k (4–8 active), so individual experts are narrow specialists. The total parameter count scales with E, but the per-token compute cost scales only with k. Gemma 4 26B-A4B [16] uses $E = 128$, $k = 8$, and 30 transformer blocks in which every block is a MoE block (additionally containing a shared dense FFN branch).

2.2 Existing Compression Approaches

Post-training quantization (PTQ). Methods such as GPTQ [5] and AWQ [11] minimize quantization error on calibration data by applying second-order weight updates (GPTQ) or activation-aware scaling (AWQ). They reduce precision uniformly across weights but do not reduce the number of experts; memory and compute reductions are proportional to the bit-width reduction alone.

Importance-based structured pruning. L0 regularization [13], movement pruning [15], and magnitude-based pruning [7] prune weight tensors or entire components (heads, layers). Applied to MoE models, structured pruning could remove entire expert networks; however, existing implementations typically require gradient access to the original training or fine-tuning pipeline, which is not available for community-distributed GGUF models.

Expert routing analysis. Several works have analyzed router behavior to understand load balancing [10] and expert specialization [4]. However, these works operate in the training regime with full gradient access. The closest antecedents to our work are expert dropping methods [1], which analyze expert importance based on routing probabilities, but do so using aggregated training statistics rather than forward-pass activation profiles on deployment calibration data.

GGUF quantization. The GGUF format [12] used by llama.cpp supports mixed-precision quantization schemes (Q4_K, Q5_K, IQ2_XXS, etc.) and an importance matrix (`imatrix`) mechanism that records per-weight activation importance for calibration-aware quantization. We repurpose this `imatrix` infrastructure for our expert selection step, treating it as a lightweight profiling instrument rather than a quantization calibration tool.

[1] Code, calibration datasets, and expert selection maps will be released on GitHub upon acceptance; replication artifacts are provided in supplementary material.

3 The Core Insight: Expert Sparsity in Deployment Domains

A MoE model trained on a global multilingual corpus must route diverse linguistic phenomena—code in dozens of programming languages, mathematics in multiple notations, natural language across hundreds of languages, and specialized domain text—across its expert pool. During training, the router learns to assign tokens from different distributions to different experts, so the full expert population is collectively necessary to serve arbitrary queries.

However, any realistic *deployment* scenario is not arbitrary. A user interacting with a model on a laptop in English, writing Python code and asking reasoning questions, is drawing on a narrow slice of the input distribution the model was trained on. The experts activated by this slice form an *empirical activation manifold* that is substantially smaller than the full expert roster.

We formalize this observation. Let $\mathcal{D}_{\text{deploy}}$ be the deployment distribution (a finitely-approximated calibration corpus) and let $C_{e,l}$ be the number of times expert e in layer l is selected during forward passes over $\mathcal{D}_{\text{deploy}}$. We define the *activation coverage* of a subset $\mathcal{S}_l \subseteq \{0, \ldots, E-1\}$ of experts in layer l as:

$$\text{Coverage}(\mathcal{S}_l) = \frac{\sum_{e \in \mathcal{S}_l} C_{e,l}}{\sum_{e=0}^{E-1} C_{e,l}}. \tag{2}$$

If the deployment distribution concentrates activation onto a small expert subset, we can retain $|\mathcal{S}_l| \ll E$ experts and still capture nearly all forward-pass activation mass. Empirically, on our calibration corpus of 1,979 tokens drawn from English programming, science, and reasoning text, retaining the top 48 of 128 experts per layer captures an average of 79.88% of activation mass across all 30 layers of Gemma 4 26B-A4B. The remaining 80 experts (62.5% of the expert pool by count) contribute only 20.12% of activation—and can be excised cleanly.

The resulting model is not a universally capable model; it is a *domain-specialist* distillation of the original. This trade-off is explicit and honest: we sacrifice breadth of generalization in exchange for throughput, memory footprint, and VRAM residency on consumer hardware.

4 Method: Empirical Activation Manifold Profiling

MAPLE proceeds in four stages: calibration corpus construction, forward-pass activation profiling, expert scoring and selection, and GGUF container assembly.

4.1 Stage 1: Calibration Corpus Construction

The calibration corpus $\mathcal{D}_{\text{deploy}}$ must be representative of the target deployment domain. For our English programming-and-reasoning target domain, we constructed a 1,979-token corpus spanning:

- Python programming tasks (Fibonacci, sorting, data structures, decorators)
- Mathematical reasoning (probability, calculus, linear algebra)
- Science explanation (Rayleigh scattering, thermodynamics, optics)
- Formal logic and syllogisms
- General English comprehension

The corpus was formatted using the Gemma 4 instruction chat template to ensure that the router encounters the same tokenization distribution as actual deployment queries.

4.2 Stage 2: Forward-Pass Activation Profiling via `llama-imatrix`

We use the `llama-imatrix` tool from llama.cpp [12] (build b11438-cbb7d52ec) to profile expert activations. During a forward pass over $\mathcal{D}_{\text{deploy}}$, `llama-imatrix` records two statistics for each expert weight tensor:

1. **Activation count** $C_{e,l}$: The number of tokens for which expert e was selected by the router in layer l.
2. **Input energy sum** $\Sigma_{e,l} \in \mathbb{R}^{d_{\text{ff}}}$: The cumulative sum of squared input activations across all $d_{\text{ff}} = 704$ input channels, summed over all tokens that activated expert e: $\Sigma_{e,l,j} = \sum_{\tau \in \mathcal{T}_{e,l}} h_{\tau,j}^2$.

These statistics are recorded into a GGUF importance matrix file (`comprehensive_imatrix.gguf`, 108.6 MB) under the keys `blk.{l}.ffn_down_exps.weight.counts` and `blk.{l}.ffn_down_exps.weight.in_sum2`.

4.3 Stage 3: Expert Scoring and Selection

For each layer l and each expert $e \in \{0, \ldots, 127\}$, we compute the MAPLE composite score:

$$s_{e,l} = C_{e,l} \cdot \sqrt{\bar{\sigma}_{e,l} + \epsilon}, \tag{3}$$

where $\bar{\sigma}_{e,l} = \frac{1}{d_{\text{ff}}} \sum_{j=1}^{d_{\text{ff}}} \Sigma_{e,l,j}$ is the mean input energy per channel for expert e in layer l, and $\epsilon = 10^{-8}$ is a numerical stability constant.

We select the top-$k' = 48$ experts per layer by score, producing a 30-entry selection map $\mathcal{S} = \{\mathcal{S}_0, \ldots, \mathcal{S}_{29}\}$ with $|\mathcal{S}_l| = 48$ for all l. The choice $k' = 48$ was driven by the VRAM budget constraint of 6141 MiB on the RTX 4050 6GB: the expert weight tensors shrink by 62.5%, yielding a total model size of 4.75 GB that fits entirely in VRAM.

4.4 Stage 4: GGUF Container Assembly

Given the selection map $\mathcal{S}$, we construct the compressed GGUF container by streaming through the source GGUF file and applying axis-0 index selection to all per-layer expert tensors:

Algorithm 1 MAPLE Container Assembly

Require: Source GGUF path P_{src}, selection map $\mathcal{S}$, target $k' = 48$
Open P_{src} as `reader`
Initialize `writer` with updated metadata: `expert_count` $\leftarrow k'$
for each tensor T in `reader.tensors` **do**
  Parse layer index l from tensor name
  if tensor name matches `ffn_gate_inp.weight` **and** $l \in \mathcal{S}$ **then**
    $T_{\text{out}} \leftarrow T[\mathcal{S}_l, :]$ ▷ Router gate weight rows for kept experts
  else if tensor name matches `ffn_down_exps.{weight, scale}` **and** $l \in \mathcal{S}$ **then**
    $T_{\text{out}} \leftarrow T[\mathcal{S}_l, \ldots]$ ▷ Expert down-proj; preserve raw quantization dtype
  else if tensor name matches `ffn_gate_up_exps.weight` **and** $l \in \mathcal{S}$ **then**
    $T_{\text{out}} \leftarrow T[\mathcal{S}_l, \ldots]$ ▷ Expert gate+up-proj; axis-0 selection
  else
    $T_{\text{out}} \leftarrow T$ ▷ All other tensors pass through unchanged
  end if
  Write T_{out} to `writer` preserving original quantization dtype
end for
Flush, close, and atomic-rename to final target path

All per-expert tensors in the source GGUF are preserved in native IQ2_XXS format without re-quantization. Scales are maintained at exactly 1.0× relative to the source.

5 Three Failure Modes and Why They Fail

5.1 Failure Mode 1: Layer-0 Embedding Proxy for Deep-Layer Routing

What was done. The first compression attempt estimated expert importance by projecting the average calibration token embedding through each layer's router gate matrix $W_{\text{gate}}^{(l)} \in \mathbb{R}^{d \times E}$ to obtain estimated routing logits: $\hat{\mathbf{g}}^{(l)} = \bar{\mathbf{e}}_{\text{calib}} W_{\text{gate}}^{(l)}$.

Why it fails. In actual inference, the routing logit at layer l is $\mathbf{h}^{(l)} W_{\text{gate}}^{(l)}$. For $l \geq 10$, hidden states have undergone multiple rounds of non-linear multi-head attention and FFN updates, rotating representations into spaces orthogonal to $\bar{\mathbf{e}}_{\text{calib}}$. The proxy misclassified 75% of active experts in deep layers, producing infinite degeneration loops (e.g., repeating `nums_nums_nums...`).

The fix. Run a full forward pass via `llama-imatrix`, ensuring expert selection reflects true hidden-state routing.

5.2 Failure Mode 2: Cascaded Scale Suppression

What was done. An intermediate attempt multiplied `ffn_down_exps.scale` by 0.795 to counteract the shrinkage of the softmax denominator when moving from 128 to 48 experts.

Why it fails. Applied across 30 consecutive transformer blocks, the suppression compounded exponentially: $0.795^{30} \approx 0.001$. The effective magnitude of the FFN contribution to the residual stream was suppressed by a factor of 1,000, reducing the model to a shallow attention passthrough devoid of factual knowledge.

The fix. Preserve all scale tensors at exactly $1.0\times$. The runtime automatically handles softmax normalization across the surviving logits.

5.3 Failure Mode 3: Sampling Hyperparameter Interference

What was done. An early interactive inference harness applied `-dry-allowed-length 2` to penalize repetitive loops.

Why it fails. Python source code depends heavily on short repeated character sequences (`def`, `if`, `in`, `+=`, `==`, `[0`). Penalizing these n-grams prevented the model from emitting valid syntax, resulting in broken pseudo-code.

The fix. Restrict sampling to temperature ($\tau = 0.4$), top-p nucleus sampling (0.9), and mild repetition penalty (1.1).

6 Experiments

6.1 Hardware and Software Configuration

Experiments were conducted on a single laptop GPU: NVIDIA GeForce RTX 4050 Laptop GPU (6141 MiB VRAM), llama.cpp build b11438, Windows 11.

6.2 Quality Benchmark

We evaluated quality using a four-task semantic fidelity benchmark testing distinct competencies (Table 1).

Table 1: Task-based semantic fidelity benchmark: compressed model (48 experts, 100% VRAM residency).

Task	Domain	Key Evaluation Terms	Matched	Fidelity	TPS
T1: Fibonacci	Code generation	`def fibonacci`, `return`, `append`, `[0,1]`	4/4	100%	45.1
T2: BST	Data structures	`class Node`, `class BST`, `def insert`, `def search`	3/4	75%	45.4
T3: Rayleigh scattering	Science reasoning	`Rayleigh`, `wavelength`, `scattering`, `blue`	4/4	100%	45.4
T4: Syllogistic logic	Formal logic	`yes`, `transitive`, `syllogism`, `lazzies`	4/4	100%	45.4
Mean				**93.8%**	**45.3**

The T2 partial score (75%) was attributable solely to token cutoff at 350 tokens, cutting off `class Node`. Relaxing to 500 tokens yielded 100% completion.

6.3 Throughput Comparison

Model	Size	GPU Layers	VRAM	Throughput
Original (128 experts)	9.92 GB	16 (partial)	6141 MiB (overflows)	6.9 TPS
MAPLE (48 experts)	4.75 GB	99 (full)	≈4,180 MiB	**45.4 TPS**
Improvement	**2.09× smaller**			**6.57× faster**

6.4 Compression Ratio and Effective Bitrate

$$\text{Effective BPW} = \frac{\text{File Size (bytes)} \times 8}{\text{Total Parameters}} = \frac{4{,}745{,}987{,}168 \times 8}{25.6 \times 10^9} \approx \mathbf{1.48}\ \text{BPW} \tag{4}$$

6.5 Limitations and Out-of-Domain Perplexity

Evaluating perplexity on a held-out, general multilingual validation text (`benchmark_val.txt`) revealed a perplexity increase from 98.9 (original) to 572.4 (MAPLE). This trade-off is fundamental: MAPLE is explicitly a domain-targeted structural distillation. Users requiring broad multilingual or multi-domain support should retain a higher expert budget ($k' \geq 64$). For edge devices where memory is strictly constrained, MAPLE provides an optimal Pareto trade-off.

7 Broader Impact Statement

MAPLE democratizes the execution of 26-billion-parameter MoE architectures on consumer laptops with 6 GB GPUs, reducing cloud dependencies and operational energy consumption. By running completely offline in VRAM, it enhances user data privacy. The technique may lead to performance degradation if applied carelessly to out-of-distribution demographic domains; practitioners must curate calibration data representative of intended users.

8 Conclusion

MAPLE demonstrates that fine-grained MoE models contain substantial structural redundancy that can be excised post-hoc. By profiling empirical forward-pass activation manifolds, we compress a 26-billion-parameter MoE model to 1.48 effective bits per parameter, unlocking 45.4 TPS inference on commodity 6 GB laptop GPUs with 93.8% domain fidelity.

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